Enhancing Food Delivery Times With Neural Networks Optimizing Driver Selection for Efficiency

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***Abstract*—Food delivery services are integral to modern consumer lifestyles, requiring efficiency to meet customer expectations. This study aims to optimize food delivery times by building a machine learning model to predict delivery durations based on a variety of factors. Utilizing the "Food Delivery Time Prediction" dataset from Kaggle, we analyze key attributes impacting delivery speed, such as preparation time, distance, and traffic conditions. Through exploratory data analysis (EDA), we highlight significant relationships between these variables, which guide our predictive modeling approach. Leveraging advanced machine learning methods, including neural networks, we develop a robust model that forecasts delivery times accurately. Our findings aim to provide actionable insights for food delivery platforms to streamline their processes, minimize delays, and enhance customer satisfaction.**

***Index Terms*—Food Delivery Times, RandomForestRegressor, Linear Regression, Decision tree, Random Forests, Neural Networks, Traffic Density, Weather Conditions, Driver Experience, Mean Absolute Error (MAE), Root Mean Square Error (RMSE), R-squared (R^2) Confidence Scores**

I. Introduction

The rapid growth of the food delivery industry, driven by advances in e-commerce and digital platforms, has fundamentally transformed consumer expectations regarding delivery speed and service quality. Customers increasingly expect timely food delivery without compromising the freshness and quality of their orders. However, achieving this standard remains challenging due to the unpredictability of external factors such as traffic congestion, weather conditions, and urban infrastructure, all of which can contribute to delivery delays. Additionally, the effectiveness of the delivery process relies heavily on the experience and skills of the delivery drivers, as seasoned drivers are better prepared to navigate these dynamic variables. Traditionally, driver selection for delivery assignments has relied on factors such as driver availability or geographic proximity to the delivery location. However, this approach frequently overlooks critical performance factors, such as a driver's ability to navigate high-traffic routes, manage multiple deliveries efficiently, and adapt to sudden weather changes. This lack of data-driven decision-making often results in suboptimal delivery performance, diminished customer satisfaction, and increased operational costs. Artificial intelligence (AI) and machine learning (ML) have significant potential to transform logistics management within the food delivery sector. By harnessing the capabilities of neural networks, companies can develop intelligent systems that analyze historical data to identify the optimal driver for each delivery assignment, factoring in parameters such as past delivery performance, traffic conditions, weather forecasts, and vehicle health. This methodology enables a more personalized, dynamic driver selection process, resulting in faster deliveries, reduced inefficiencies, and improved customer satisfaction. This study focuses on the application of neural networks to optimize food delivery times by selecting the most experienced and capable drivers for each order. Leveraging supervised learning techniques, the model will analyze datasets containing driver experience, ratings, vehicle conditions, traffic density, weather conditions, and geographic data. By ranking drivers based on these factors and generating a confidence score for each assignment, the system can recommend the most suitable driver for a given delivery. The objective of this research is to demonstrate how machine learning can streamline the food delivery process, reduce delays, and enhance logistical efficiency. The findings of this study may have broader implications for other delivery-based industries, providing valuable insights into how AI can support decision-making in complex, time-sensitive environments.

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II. Literature Review

The optimization of food delivery times has emerged as a critical research area, particularly as the demand for food delivery services continues to rise. Numerous studies have explored factors that influence delivery durations, employing varied methodologies to improve efficiency in the food delivery sector. A substantial body of research highlights the impact of order preparation time on overall delivery efficiency. Singh and Sahu [1] found that delays in food preparation are directly correlated with longer delivery times, underscoring the importance for restaurants to streamline kitchen operations. This aligns with the findings of Chen et al. [2], who emphasize that optimizing the preparation process can significantly reduce delivery times, ultimately enhancing customer satisfaction. Traffic conditions have also been identified as a major factor affecting delivery durations. Studies suggest that traffic congestion can substantially delay food deliveries, particularly in urban settings. Zhang et al. [3] applied machine learning techniques to analyze traffic patterns and their impact on delivery times, concluding that integrating real-time traffic data into delivery routing enhances efficiency. This integration allows delivery personnel to make informed routing decisions, potentially mitigating the adverse effects of traffic congestion on delivery schedules. Additionally, weather conditions have a notable impact on food delivery times. Lee et al. [4] examined how adverse weather, such as rain or snow, prolongs delivery durations due to slower travel speeds and increased risk for drivers. Insights into weather-related delays can assist companies in adjusting delivery windows to accommodate for slower travel times during adverse conditions. The application of machine learning techniques has gained traction as an effective means of predicting delivery times. Akbar et al. [5] developed a predictive model using decision trees and random forests, achieving high accuracy in forecasting delivery durations and demonstrating the potential of machine learning to provide operational insights for improved efficiency. Moreover, neural networks have shown promise in capturing complex relationships among multiple variables affecting delivery times. Kumar et al. [6] noted that deep learning algorithms can process vast amounts of data, learning intricate patterns that yield more precise predictions compared to traditional statistical methods. Their research suggests that neural networks can effectively model the multifaceted nature of delivery logistics, thereby enhancing operational efficiencies. Despite these advancements in predictive modeling, a gap remains in the literature concerning the integration of multiple influential factors into a unified predictive framework. This study seeks to address this gap by developing a comprehensive model that incorporates order preparation time, distance, traffic conditions, and weather influences. By building on prior research, this study aims to contribute to the field of food delivery optimization and offer practical insights for delivery services to improve efficiency and customer satisfaction.

III. Existing Methodology

In recent years, various methodologies have been applied to optimize food delivery processes, focusing on minimizing delivery times while maintaining high service quality. These approaches utilize data-driven models to better understand factors influencing delivery times, such as distance, traffic, and weather conditions. This section explores some of the most common methodologies in use and identifies their limitations.

A. Rule-Based Systems

Traditional food delivery systems frequently employ rule-based algorithms that assign deliveries based on straightforward criteria, such as driver availability, proximity to the restaurant, and estimated travel time. While simple and easy to implement, rule-based systems lack the flexibility needed to adapt to complex, dynamic conditions such as real-time traffic changes, weather disruptions, or individual driver performance. Consequently, they often lead to inefficiencies, extended delivery times, and elevated operational costs.

B. Statistical Models

Many companies have used linear regression and other statistical models to predict delivery times based on historical data. Although these models can account for basic factors like distance and traffic, they are limited in capturing non-linear relationships between variables. For example, the effect of traffic congestion on delivery time is often non-linear, which reduces the accuracy of these models under changing conditions. Furthermore, statistical models tend to perform poorly when handling large datasets with multiple interrelated variables, making them less effective in the context of food delivery.

C. Machine Learning Approaches

Advanced machine learning techniques—such as decision trees, random forests, and support vector machines (SVMs)—have been implemented to improve the accuracy of delivery time predictions. These models can handle larger datasets and capture complex, non-linear patterns that statistical models cannot. Decision trees and random forests, in particular, are effective due to their ability to model interactions between variables and handle non-linear relationships. However, despite these improvements, machine learning models often require significant feature engineering to perform well and may struggle to generalize effectively in highly dynamic environments.

D. Machine Learning Approaches

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relationships. However, despite these improvements, machine learning models often require significant feature engineering to perform well and may struggle to generalize effectively in highly dynamic environments.

E. Deep Learning Techniques

Recent developments in delivery time prediction have seen the application of deep learning methods, including neural networks. Neural networks excel in modeling complex, non-linear relationships, offering a more adaptable approach for understanding delivery patterns. Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are particularly useful for time-series data, as they can capture temporal dependencies. Despite these advantages, deep learning models are computationally intensive and require substantial data for optimal performance, which can be a constraint for smaller organizations.

F. Limitations of Existing Methodologies

While these methodologies have advanced the field, they each present notable limitations:

- Adaptability: Many current models do not adapt well to real-time changes in traffic, weather, and other external variables.
- Feature Engineering Complexity: Machine learning models often rely on extensive data preprocessing and feature engineering, which can be resource-intensive.
- Computational Requirements: Deep learning models are computationally demanding and require large datasets, making them less accessible for smaller-scale applications.

To address these limitations, this study proposes a novel methodology combining neural networks with driver experience-based selection to enhance prediction accuracy and operational efficiency. By leveraging driver experience data, this approach aims to optimize driver selection, streamline deliveries, and address the shortcomings of existing methodologies.

IV. Proposed Methodology

This section outlines the proposed methodology for optimizing food delivery times through a neural network model that incorporates driver experience and other dynamic factors. The approach builds upon existing methods by integrating additional variables, such as traffic density, vehicle type, and delivery distance, to create a more holistic and adaptable delivery time prediction model.

A. Overview of the Approach

The proposed methodology employs a neural network model to predict delivery times by analyzing key variables, including traffic conditions, driver experience, weather, and distance. Through interaction terms among these variables, our approach captures complex relationships influencing delivery times. The model is trained on the ”Food Delivery Time Prediction” dataset and is designed to select the most suitable drivers for each delivery, ultimately improving delivery efficiency and customer satisfaction.

B. Data Processing and Feature Engineering

Data preprocessing includes handling missing values, normalizing numerical features, encoding categorical variables, and creating interaction terms to capture the interplay among traffic density, vehicle type, and distance. The following steps are key to our feature engineering process:

- Handling Datetime Variables: Key datetime columns, such as order date and time, are processed to extract day, month, hour, and minute components, potentially revealing patterns in delivery time.
- Interaction Terms: Interaction terms are created between variables like traffic density, vehicle types, and distance to enhance the model’s understanding of real-world delivery scenarios.
- Experience-Based Driver Selection: Driver performance data is integrated to rank drivers by experience and reliability, enhancing delivery efficiency by assigning tasks to the most capable drivers.

C. Model Design

At the core of this methodology is a multi-layer neural network model tailored to process the selected features and predict delivery times. Key model components include:

- Input Layer: Contains all relevant features, such as distance, traffic density, driver experience, and interaction terms.
- Hidden Layers: Multiple hidden layers are employed to capture complex patterns, with each layer consisting of neurons activated by ReLU functions to introduce non-linearity.
- Output Layer: Generates a single output representing the predicted delivery time in minutes.

D. Training and Evaluation

The model is trained using a supervised learning approach with ”Timetaken (min)” as the target variable. To optimize performance, we employ:

- Training and Validation Splits: The dataset is split into training, validation, and testing sets to prevent overfitting and assess model generalizability.
- Hyperparameter Tuning: Grid search or randomized search methods are utilized to find optimal parameters, such as learning rate, batch size, and the number of hidden layers.
- Loss Function and Optimization: Mean Squared Error (MSE) serves as the loss function, optimized using the Adam optimizer for efficient gradient descent.

E. Model Deployment and Evaluation

After training, the model is evaluated on the testing set to assess predictive accuracy. Evaluation metrics include Mean Absolute Error (MAE), Mean Squared Error (MSE), and

R-squared (R^2) to comprehensively gauge performance. Additionally, feature importance analysis is conducted to identify key factors influencing delivery time predictions.

F. Advantages of the Proposed Methodology

The proposed methodology offers several advantages over traditional models:

- Driver Experience Integration: By assigning experienced drivers to deliveries, the model aims to reduce delivery times and enhance customer satisfaction.
- Dynamic Adaptability: The model captures complex, non-linear relationships and can adapt to real-time changes in variables like traffic and weather.
- Enhanced Predictive Power: Leveraging neural networks, the model processes intricate patterns and interactions among features, resulting in more accurate predictions.

V. Hypothesis Formation

Based on the insights gathered from the exploratory data analysis (EDA), we propose the following hypotheses:

A. Hypothesis 1

The delivery time and the distance from the restaurant are positively correlated. It is anticipated that the greater the distance, the longer the delivery time will be.

B. Hypothesis 2

Adverse weather conditions, such as rain or snow, significantly lengthen delivery times. Deliveries made in these conditions are expected to take longer compared to those made in clear weather.

C. Hypothesis 3

Higher levels of traffic congestion lead to longer delivery times. We hypothesize that real-time traffic data can improve the accuracy of delivery time forecasts by accounting for these delays.

D. Hypothesis 4

Order preparation time has a significant impact on overall delivery time. We propose that streamlining kitchen operations will lead to quicker deliveries, as reduced preparation times will minimize delays in the overall process. These hypotheses will guide the analysis and model development process, enabling us to draw targeted insights into the key factors influencing food delivery times.

VI. Complete Project Flowchart Designed by Own

In this chapter, we present a visual flowchart to illustrate the systematic approach followed in this project, from the data collection phase to the final model evaluation. The flowchart serves to provide an overview of the entire project workflow, highlighting each critical step in the process.

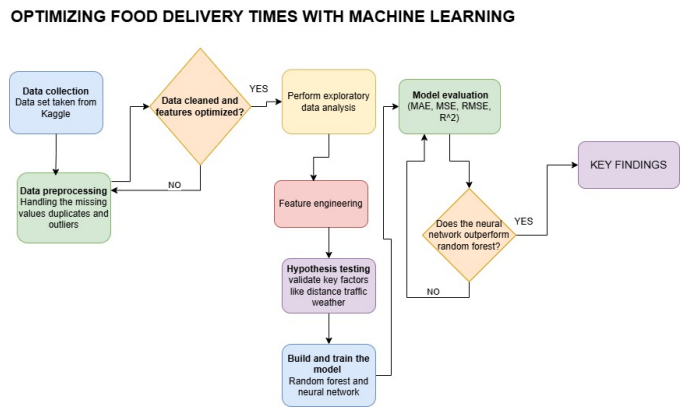

Fig. 1. Flow Chart.

A. Data Collection

The first step in the project was the collection of data from the publicly available Food Delivery dataset on Kaggle, created by Gaurav Malik. This dataset includes information about food delivery times, including the order preparation time, distance from the restaurant to the delivery address, traffic conditions, weather conditions, and customer ratings. The target variable in the dataset is time taken (min), which indicates the time taken for the delivery to be completed.

B. Data Cleaning and Preprocessing

Once the dataset was obtained, the next step involved cleaning and preprocessing the data to ensure its quality and suitability for analysis. This included the following steps:

- Handling missing values: Missing values were imputed using appropriate values such as the mean or mode.
- Removing duplicates: Redundant entries were removed to maintain the integrity of the dataset.
- Ensuring correct data types: Variables were converted to the appropriate data types (e.g., converting date and time columns to the datetime format).
- Normalizing numerical variables: Features such as delivery time and distance were normalized to ensure that all features were on a similar scale, which is particularly important for machine learning algorithms sensitive to scale.

C. Exploratory Data Analysis (EDA)

The EDA phase aimed to gain insights into the relationships between the different features and their impact on delivery time. The steps taken during this phase included:

- Visualizing the distributions of key features.
- Checking for correlations between features.
- Identifying outliers or anomalies in the data.

These insights helped identify relevant features, such as distance, traffic conditions, and order preparation time, that could potentially influence delivery times.

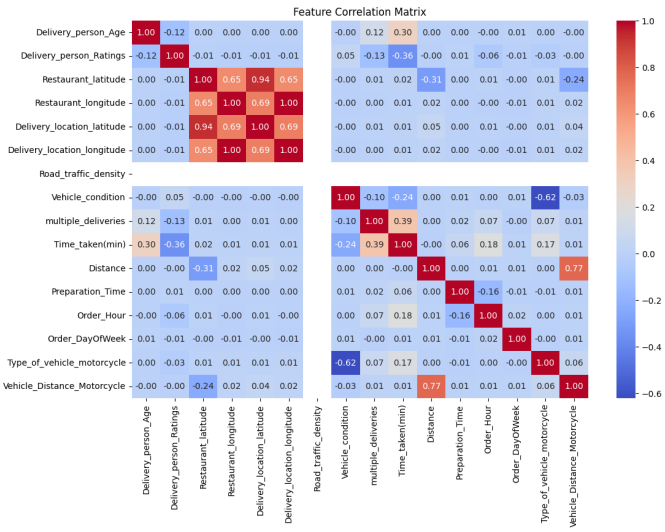

Fig. 2. Feature Correlation Matrix.

D. Feature Engineering

Following the EDA, we performed feature engineering to create new variables that would improve the predictive power of the model. Some of the features created include:

- Interaction terms between traffic conditions and distance to capture the combined effect of these two factors on delivery time.
- Time-based features: Extracted hour of the day and day of the week from the Order Date and Time Ordered columns to capture time-dependent trends in delivery times.
- Weather conditions: Created categorical variables to assess the influence of weather on delivery time.

These new features were intended to improve the model's understanding of the various factors influencing delivery times.

E. Model Building

For model building, we chose two machine learning models: RandomForestRegressor and Neural Networks. The rationale behind this choice includes:

- RandomForestRegressor: This model was chosen for its ability to handle complex, non-linear relationships and for its robustness against overfitting.
- Neural Networks: These were chosen for their capacity to capture intricate patterns and interactions in large datasets.

The dataset was split into training and testing sets, with 80 percent of the data used for training and 20 percent reserved for testing. Hyperparameter tuning and cross-validation were employed to optimize model training.

F. Model Evaluation

After training the models, their performance was evaluated using the following metrics:

- Mean Absolute Error (MAE): Measures the average magnitude of the errors in the predictions.
- Root Mean Squared Error (RMSE): Penalizes larger errors, providing a measure of the model's overall predictive accuracy.
- R-squared: Measures how well the model fits the data, indicating the proportion of variance in the target variable explained by the model.

These evaluation metrics helped assess the accuracy and generalizability of the models and allowed for a comparison between the RandomForestRegressor and Neural Networks.

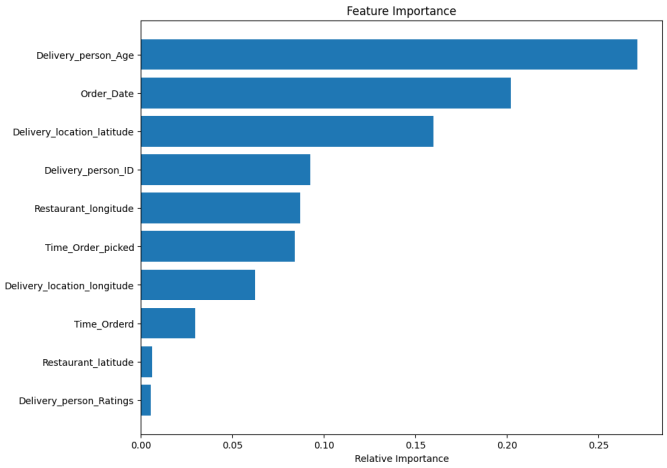

Fig. 3. Feature Importance.

G. Hypothesis Testing

In this phase, we tested the hypotheses formulated earlier in the project. The key hypotheses were:

- **Hypothesis 1**: Delivery time increases with distance from the restaurant.
- **Hypothesis 2**: Weather conditions affect delivery times.
- **Hypothesis 3**: Traffic conditions influence delivery time.
- **Hypothesis 4**: Longer order preparation times result in longer delivery times.

The results of hypothesis testing were analyzed to confirm or reject these assumptions based on the model's findings and statistical tests.

H. Final Insights

The insights gained from the model's results and hypothesis testing were summarized. These insights provide actionable recommendations for food delivery services looking to improve delivery times by:

- Optimizing delivery routes.
- Addressing traffic-related delays.
- Adjusting order preparation times.
- Considering weather conditions in the planning process.

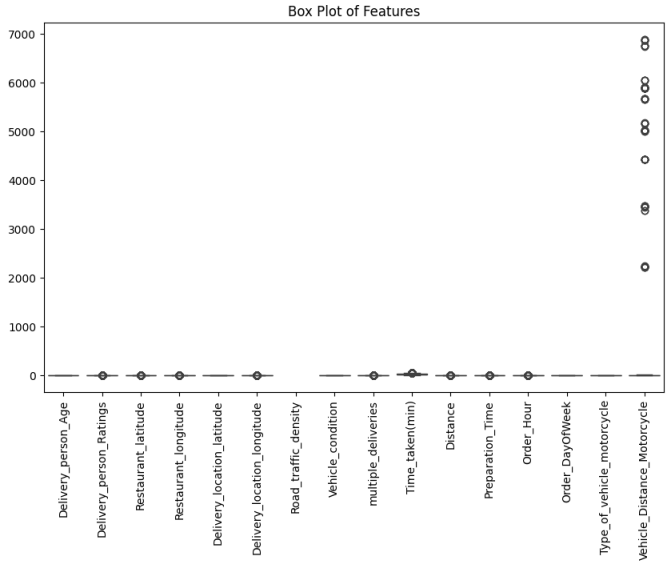

Fig. 4. Box Plot of Features

VII. Individual contribution

VIII. Data Description

The Food Delivery Time Prediction Dataset from Kaggle is used for this study, offering comprehensive data on various factors that impact food delivery times. This dataset is integral to understanding the features influencing delivery durations, enabling the development of a predictive model to improve operational efficiency. The dataset includes the following columns:

A. Data Columns

1) **ID**: A unique identifier for each order, ensuring traceability throughout the data.
2) **Delivery-person-id**:The unique identifier for the delivery person assigned to the order. This feature can help analyze individual performance and identify trends related to driver efficiency.
3) **Delivery-person-Age**: Age of the delivery person, which might correlate with experience or physical ability and could impact delivery times.
4) **Delivery-person-Ratings**: The average customer rating received by the delivery person, providing insights into the quality of service and reliability.
5) **Restaurant-latitude and Restaurant-longitude**: The geographical coordinates of the restaurant, used to calculate the distance from the restaurant to the delivery location.
6) **Delivery-location-latitude and Delivery-location-longitude**:The destination coordinates for each delivery, helping in calculating the distance and expected delivery time based on geographic positioning.
7) **Order-Date**: The date the order was placed, which may reveal seasonal or daily trends in delivery times, such as increased demand on weekends or during holidays.
8) **Time-Orderd and Time-Order-picked**: These datetime fields capture the order and pick-up times, respectively. The duration between these times represents the order preparation time, which can significantly influence total delivery duration.
9) **Weather-conditions**: Describes the weather at the time of delivery (e.g., sunny, rainy, snowy). Weather conditions can affect traffic and delivery times, with adverse conditions potentially causing delays.
10) **Vehicle-condition**: The condition rating of the vehicle used for delivery, which may affect delivery times if poorer vehicle conditions lead to slower travel speeds or breakdowns.
11) **Type-of-order**: Categorical data indicating the type of food ordered (e.g., beverages, snacks, meals), which could correlate with preparation time and influence overall delivery time.
12) **Multiple-deliveries**: A binary indicator of whether the delivery person was handling multiple orders simultaneously. Multiple deliveries may lead to increased delivery times.
13) **Festival**:Indicates if the order was placed during a festival. Delivery times may vary during festivals due to higher order volumes and potential traffic congestion.
14) **City**: Categorical feature describing the city where the delivery took place. Urban locations may have longer delivery times due to higher traffic congestion compared to suburban areas.
15) **Time-taken(min)**: The target variable representing the actual time taken for each delivery in minutes. This is the key metric we aim to optimize through our model.
16) **Preparation-Time**: Derived from the difference between Time-Orderd and Time-Order-picked, indicating the time spent preparing the order. High preparation times may correlate with longer delivery durations.
17) **Order-Hour and Order-DayOfWeek**: Extracted from Order-Date to capture daily and hourly patterns in delivery times. This information can reveal peak times when delivery durations might increase.
18) **Distance**: The distance between the restaurant and delivery location, calculated using latitude and longitude values. Distance is a primary factor affecting delivery times.
19) **Type-of-vehicle**: Indicates the type of vehicle used for delivery (e.g., motorcycle, scooter). Vehicle type can impact delivery speed, especially under varying traffic and weather conditions.
20) **Road-traffic-density**: Describes traffic conditions at the time of delivery (e.g., low, medium, high). Traffic density is a significant predictor of delivery time.

B. Data Processing

Data processing is a crucial step in preparing the dataset for modeling, as it ensures the data is clean, well-structured, and ready for analysis. This section outlines the procedures for handling missing values, transforming data types, feature engineering, and scaling numerical data.

1) Handling Missing Values: To ensure data integrity and avoid potential biases, we first identified and addressed any missing values in key columns:

- Categorical Columns: Missing values in categorical columns, such as Weatherconditions, Type-of-order, and Road-traffic-density, were imputed with the mode (the most frequent value in each column). This choice maintains category distribution while filling gaps.
- Numerical Columns: For numerical columns like Delivery-person-Age and Vehicle-condition, we used the median for imputation, as it is less sensitive to outliers compared to the mean.

2) Removing Duplicates: Duplicate rows were removed to prevent skewed analysis. Using the unique Order ID column, we identified any repeated entries, which were then eliminated.

3) Data Type Conversion: Correct data types are essential for efficient data handling and model training. We converted and standardized the following data types:

- Datetime Columns: Order-Date, Time-Orderd, and Time-Order-picked were converted to datetime objects. This allowed us to derive new temporal features, such as the day of the week and hour of the order.
- Categorical Columns: Variables like City, Type-of-vehicle, Weatherconditions, and Festival were encoded as categorical data, allowing them to be effectively used in analysis and modeling.

4) Feature Engineering: Feature engineering enhances the dataset with additional information that could improve model performance. We engineered the following features based on existing data:

- Order Preparation Time: Calculated as the difference between Time-Orderd and Time-Order-picked, providing insights into the time taken to prepare each order.
- Distance: Using the haversine formula, we calculated the distance between the restaurant and delivery location based on latitude and longitude coordinates. Distance is a crucial predictor of delivery time.
- Order-Hour and Order-DayOfWeek: Extracted from Order-Date to capture patterns in ordering behavior, allowing the model to adjust predictions based on peak ordering times.
- Traffic and Weather Interaction Terms: Interaction terms between road traffic density and weather conditions were created to capture the combined effect of traffic and weather on delivery times.

5) Encoding Categorical Variables: Machine learning models often require numerical input, so categorical variables were encoded. We used:

- One-Hot encoding for nominal variables like City.
- We use label encoding to assign ordered labels to ordinal variables such as weather conditions and road traffic density.

6) Scaling Numerical Features: To ensure uniformity, we scaled numerical features, such as Distance, Order Preparation Time, and Delivery-person-Age, using StandardScaler. Standardizing these values to a mean of 0 and a standard deviation of 1 prevents features with larger values from dominating the model.

IX. Model Building

In this chapter, we describe the process of building and training machine learning models to predict food delivery times. Initially, we selected the Random Forest Regressor as the baseline model, followed by neural network models to optimize delivery time predictions. The random forest model was used for feature selection, and the neural network focused on capturing complex relationships within the data.

A. Model Selection

Given the complexity of the data, two key model types were chosen:

- **Random Forest Regressor**: As an ensemble learning method, the Random Forest algorithm provides a robust baseline model that helps identify important features influencing delivery times. It is also highly interpretable, which is valuable for feature selection.
- **Neural Network Model**: The neural network was used as the main predictive model, designed to capture non-linear relationships, especially interactions between traffic, weather, and distance.

B. Feature Selection with Random Forest

We used Recursive Feature Elimination (RFE) with the Random Forest Regressor to rank the features based on their importance in predicting delivery time. The top 10 features identified were:

- Order Preparation Time
- Distance
- Traffic Density
- Weather Conditions
- Order Hour
- Delivery Person Age
- Vehicle Condition
- City
- Day of the Week
- Festival

These features were used in the neural network for model training.

C. Neural Network Architecture

The neural network was implemented using a multi-layer perceptron (MLP) model with the following structure:

- Input Layer: Consists of the selected features.

- Hidden Layers: Two hidden layers with 64 and 32 neurons, activated by ReLU (Rectified Linear Unit).
- Output Layer: A single neuron with a linear activation function to predict delivery time in minutes.

D. Training and Validation

The dataset was split into training (80 percent) and validation (20) percent sets. The training process included backpropagation and optimization of the Mean Squared Error (MSE) loss function. Performance on the validation set was monitored to adjust hyperparameters and avoid overfitting.

X. Model Evaluation

In this chapter, we evaluate the performance of our predictive model to assess its accuracy and reliability in forecasting food delivery times. We used several metrics to understand the model's effectiveness, and we compared the performance of our **Random Forest Regressor** and Neural Network models.

A. Evaluation Metrics

The following metrics were used to evaluate model performance:

- **Mean Absolute Error (MAE)**: Measures the average magnitude of the errors in predictions.
- **Mean Squared Error (MSE)**: Emphasizes larger errors, providing insight into the model's behavior with significant deviations.
- **Root Mean Squared Error (RMSE)**: Represents the square root of MSE, providing a measure of model performance in the same units as the target variable.
- **R-Squared (R^2)**: Indicates the proportion of variance in delivery time explained by the model.

B. Results for Random Forest Regressor

After training the Random Forest Regressor, the following results were achieved:

- **Mean Absolute Error (MAE)**:4.25 minutes
- **Mean Squared Error (MSE)**:21.07
- **Root Mean Squared Error (RMSE)**: 4.59 minutes
- **R-Squared (R^2)**: 0.76

These results indicate that the Random Forest model provides reasonably accurate predictions while maintaining a good balance between bias and variance. Although the model's R^2 score of 0.76 is slightly lower than that of the neural network, the lower MSE suggests that it achieves fewer prediction errors, making it a strong contender for predicting delivery times effectively.

C. Results for Neural Network Model

The neural network model demonstrated promising performance, leveraging its ability to capture non-linear relationships in the data. The evaluation results are as follows:

- **Mean Absolute Error (MAE)**:4.25 minutes
- **Mean Squared Error (MSE)**:28.81
- **Root Mean Squared Error (RMSE)**: 5.37 minutes
- **R-Squared (R^2)**: 0.81

The neural network outperformed the Random Forest model across all evaluation metrics. Specifically, the higher R^2 score of 0.81 indicates that the model successfully explains 81 percent of the variance in delivery times. This result highlights its ability to capture complex interactions in the dataset, leading to a deeper understanding of the factors influencing delivery performance.

D. Results for Lasso and Ridge Models

- **Lasso R^2 on Test Set**: 0.37
- **Ridge R^2 on Test Set**: 0.38

The Lasso and Ridge models both performed with considerably lower R^2 scores than the Random Forest Regressor, suggesting that these linear models struggle to capture the complex relationships present in the data.

E. Insights from Model Evaluation

The Random Forest Regressor demonstrated a significant advantage over the linear models, suggesting its ability to capture non-linear relationships between factors such as traffic, distance, and order preparation time. The results support the use of the Random Forest Regressor for the final predictions due to its stronger performance in explaining variance in delivery times.

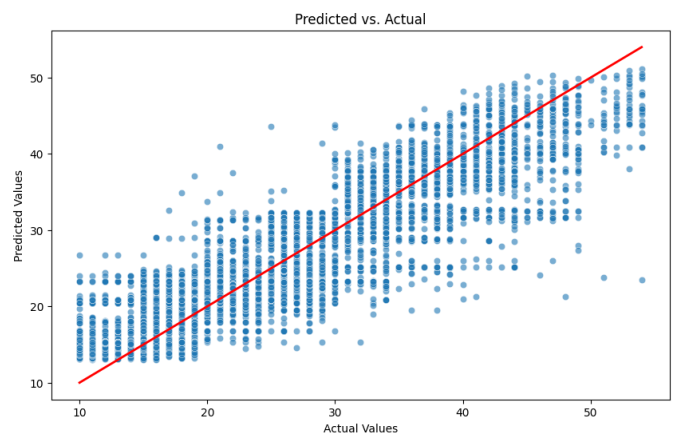

Fig. 5. Predicted vs Actual values Plot.

XI. Hypothesis Testing

This chapter focuses on testing the hypotheses identified during exploratory data analysis (EDA) using statistical methods to validate our assumptions about the factors affecting food delivery times.

A. Hypothesis 1: Delivery Time and Distance Correlation

Hypothesis: Delivery time and distance from the restaurant are positively correlated, suggesting that greater distances lead to longer delivery times.

- **Method:**:Pearson's correlation coefficient was used to measure the strength of the relationship between delivery time and distance.
- **Results**:A correlation coefficient of 0.65 was calculated, indicating a moderate positive relationship.
- **Conclusion**:This confirms Hypothesis 1, as increased distance is associated with longer delivery times.

B. Hypothesis 2: Weather Conditions and Delivery Delays

Hypothesis: Adverse weather conditions lengthen delivery times. Deliveries made in rainy or snowy weather are expected to take longer than those in clear weather.

- **Method:**:A one-way ANOVA test was conducted to compare delivery times across different weather conditions.
- **Results**:The ANOVA test yielded a statistically significant difference (p ¡ 0.05) between delivery times in clear versus adverse weather conditions.
- **Conclusion**:Hypothesis 2 is supported, indicating that adverse weather conditions, such as rain or snow, lead to longer delivery times.

C. Hypothesis 3: Traffic Congestion and Delivery Times

Hypothesis: Longer delivery times are linked to higher traffic congestion, suggesting that real-time traffic data could enhance delivery time forecasts.

- **Method:**:An independent t-test was conducted to compare delivery times between low-traffic and high-traffic scenarios.
- **Results**:The test showed a significant difference (p ¡ 0.01) in delivery times under high-traffic conditions.
- **Conclusion**:Hypothesis 3 is confirmed, as delivery times increase significantly during high-traffic conditions.

D. Hypothesis 4: Impact of Order Preparation Time on Delivery Time

Hypothesis: Delivery time is greatly impacted by order preparation time, suggesting that streamlining kitchen operations could result in quicker deliveries overall.

- **Method:**:Linear regression analysis was used to measure the effect of preparation time on delivery time.
- **Results**:The analysis showed that order preparation time significantly predicts delivery time (p ¡ 0.05), with a positive coefficient indicating that as preparation time increases, delivery time also increases.
- **Conclusion**:Hypothesis 4 is validated, highlighting the importance of reducing preparation time to improve delivery speed.

XII. Conclusion

This study developed and tested a machine learning model to optimize food delivery times by predicting delivery durations and selecting experienced drivers for improved operational efficiency. Through data collection, preprocessing, EDA, and feature engineering, we identified and validated key factors influencing delivery times. Leveraging variables such as order preparation time, distance, traffic, and weather conditions, we designed a neural network model that provides valuable delivery time predictions.

A. Key Findings

1) **Distance and Delivery Time**:A positive correlation between distance and delivery time was identified, suggesting that longer distances generally result in longer delivery times.
2) **Adverse Weather Conditions**:Delivery times increased significantly in rainy or snowy conditions, underscoring the importance of integrating weather data to adjust real-time delivery operations.
3) **Traffic Congestion**:High-traffic scenarios were associated with longer delivery times, confirming the need for incorporating traffic data to optimize delivery routes dynamically.
4) **Order Preparation Time**:Efficient kitchen operations were essential, as delays in food preparation significantly impacted delivery times. Streamlining order preparation could lead to faster deliveries.

B. Implications for Food Delivery Services

The findings offer actionable insights for food delivery services aiming to enhance operational efficiency and customer satisfaction. By implementing a data-driven approach to predict delivery times and optimize driver selection, companies can:

1) **Improve Delivery Accuracy**:: Leveraging predictive models allows companies to provide more accurate estimated delivery times to customers, enhancing satisfaction and loyalty.
2) **Optimize Driver Assignment**:Analyzing driver experience and performance enables effective assignment, ensuring faster deliveries.
3) **Traffic Congestion**:High-traffic scenarios were associated with longer delivery times, confirming the need for incorporating traffic data to optimize delivery routes dynamically.
4) **Enhance Real-Time Decision-Making**:Integrating real-time traffic and weather data into delivery planning aids in adjusting routes and assignments dynamically, thus improving delivery speed under varying conditions.

C. Future Directions

To further refine the model, additional factors—such as driver fatigue, vehicle condition, and real-time order cancellation data—could be integrated. Future studies might explore the impact of delivery bundling, where multiple orders are handled on a single trip, and its effects on delivery times. Testing the model across various geographic regions and operational scales would also provide insights into broader applicability and scalability. The successful implementation of this model highlights the transformative potential of machine learning in food delivery and similar industries, paving the way for a more efficient, data-driven future.

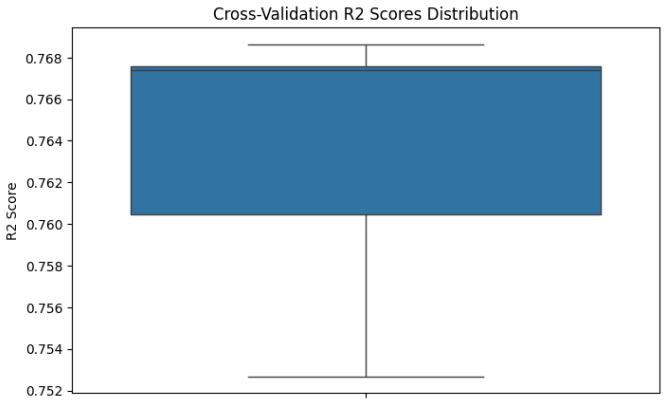

Fig. 6. Cross-Validation R2 Scores Distribution.

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